

SPOTIFY CHARTS ANALYSIS 2017 - 2018

By Anuraag v.p.

Tools : Pandas , SQL , Power BI

Project Overview :

The project analyses two years of real day to day activity of tracks performing on Top200 and Viral50 charts and how music moved through the years of 2017 and 2018 . These charts are published daily across 66 countries in the chosen dataset . Rather than working on a pre aggregated dataset , from a single massive csv file [3.5 GB] and build a proper relational based database . The cleaning and filtering were done in python via Pandas , preparing the star schema and analysis was done on SQL Server , and preparing visualization was done on Power BI . The pipeline was chosen for better performing analysis workflow , rather than a single dataset exercise .

The 2017 -18 time period was deliberately chosen because I believe that it was the peak of most genres of music such as Hip Hop , Pop , Pop ã rap , Rock etc. and it genuinely captured an inflection point in pop music including the commercial peak of the infamous SoundCloud Era Rap ã. Several of hip-hop's most memorable Albums were released during this time such as Astroworld by Travis Scott , Scorpion by Drake , Beer bongs & Bentley by Post Malone and etc.

Data source and Methodology :

Data Source :

The dataset used for analysis comes from Spotify charts ãpublished by dhruvildave on Kaggle . The raw file contains 26 million rows spanning from January 1 2017 to December 31 2021 . Each row contains daily entry of tracks in the Top200 and Viral50 charts and the trend movements . The Top200 chart is ranked by stream volume and Viral50 is ranked by sharing velocity on social media platforms . This project filters the raw file down to 2017-18 specifically to focus on those two years for analysis .

Cleaning (Python) :

The raw file is read in 500,000 row chunks since the unfiltered csv file is 3.5 GB . Each chunk is filtered to 2017 ÷ 2018 before being concatenated , so the working data in memory never exceeds the size of what is actually needed downstream .

A real Spotify track ID is extracted directly from row chart's URL , rather than relying on song titles as Key ID . This decision matters a lot because track titles aren't reliable identifiers , since the same underlying song can appear under near identical titles such as a whitespace difference . Hence , extracted Spotify ID becomes a stable and concrete identifiers and this will further go on to be primary key for tracks dimension table .

Schema Design and Analysis :

The cleaned data is then loaded into a staging table and normalized into a star schema . A single fact table (fact_chart_table) recording one row per chart position , per day , per region joined to three dimension tables : dim_tracks , dim_regions , dim_artists .

Two schema tables creations were date driven . Firstly , dim_artists table and dim_track because their name fields required unusually wide text columns: the longest artist credit in the dataset runs 788 characters (a various-artists compilation crediting more than a dozen individual vocalists), and the longest track title runs 511 characters. Second, dim_artists could not carry a UNIQUE constraint on artist name as originally planned .

Data Overview :

Dataset scale :

Metric	Value
Total entries ; 2017 -18	9,264,161
Regions covered	66

Rank distribution :

Statistics	Values
Count	9264161
Mean	78.97
Std. deviation	58.66
Min	1
25%	28
Median	63
75%	128
Max	200

INSIGHTS :

a] Peak ï Rank Leaderboard

For each track , the query finds the single best (lowest ï numbered) rank ever achieved in 2018 , using total chart appearances as a tie breaker . This demonstrates track's absolute peak from its overall staying power .

FINDINGS :

“God’s Plan” by Drake lead the 2018 with 20,265 chart appearances after reaching #1 at least in 1 region . Over more than 1,000 less than the top track comes “D&GAF” by Dua Lipa in #2 place . These numbers suggest an overall global presence .

b] Top artists with the most distinct tracks charting in 2018 :

Rather than counting raw chart appearances which would overweight artists whose songs simply charted in many regions at once , this query counts each artist's number of distinct charting tracks (COUNT DISTINCT track_id). This isolates genuine catalog breadth from regional reach.

FINDINGS :

Imagine Dragons and Kim Larsen solidified the #1 and #2 spot with 71 and 70 distinct tracks charting in 2018 , again Drake showed up in the top 5 list with #3 spot with 63 distinct tracks charting in 2018 . Many other massive international artists such as BTS [#7] and Avicii [#9] were in top 10 lists .

c] Regional spread : how many distinct regions did each 2018 track chart in?

The query finds out the track name and artist which charted in the highest number of regions in the year of 2018 , with the chart appearances as the tie breaker .

FINDINGS :

Dua Lipa's "DGA" triumphs at #1 with 19,235 chart appearances with charting consistently in 66 regions . Furthermore , "Havana" by Camilla Cabello , "Like It or Love It" by Cardi B , "Sad!" by XXXTentation took #2 ,#3 and #4 spots.

d] TOP200 Vs Viral50 entry volume and distinct tracks by charts type in 2018

Despite holding only 50 daily slots per region in comparison to Top 200's 200 spots , the Viral 50 chart cycles through more distinct tracks overall . 35,736 versus 27,575. This showcases a real behavioral difference between the two charts: Top 200 tends to be held by fewer, longer-running tracks, while Viral 50 turns over faster, with songs spiking and dropping off more quickly.

Chart type	Total entries	Distinct tracks
Top200	3739321	27575
Viral50	1123695	35736

e] Regional footprint : total chart entries per region in 2018

The above created table should be a useful metric for this insight with 200 and 50 slots available per day, a region with complete daily coverage across all of 2018 would show exactly $365 \times 250 = 91,250$ total entries. Twenty-eight of the 66 regions in this dataset hit that figure , demonstrating gap free chart publishing for the full year.

By contrast, four regions , which are Saudi Arabia, Egypt, and Morocco (2,150 entries each) and the United Arab Emirates (2,350) sit dramatically below every other market, at roughly 2% of the coverage ceiling. This is not a data collection gap . Spotify's service did not launch in the Middle East and North Africa region until November 2018, so for roughly the first ten months of the year there was, quite literally, no chart to record in these markets. The chart data independently corroborates Spotify's real-world regional expansion timeline.

FINDINGS :

The top 3 spots were secured by Finland , New Zealand and Italy all with precisely 91,250 appearances . Finland took the 1st place with most number of distinct tracks appeared in charts [3874] . The bottom 3 place were taken by Saudi Arabia , Egypt , Morocco with only 2150 appearances .

f] Longest MOVE_UP streak before peak, per track/region (2018, top200 only)

This analysis is mainly aimed at measuring the momentum of the track throughout the charts .How many consecutive times a track moved up in the ranks before reaching its peak rank . A classic SQL technique ñgaps and islandsòwas implemented here . A track that climbs steadily for two weeks tells a meaningfully different story than one that jumps straight to the top overnight, even if their final peak rank is identical.

FINDINGS :

The track with the highest move up was ñSerenata Cruel ñby Villa canno in Chile with the highest streak length of 221 . Second place was taken by the song ñTengo Que Colgar ñby Banda MS de Sergio Lizrraga in Mexico with streak length of 218 , barely missing the first spot by only 3 move ups .

g] 'Album bomb' days ; Artists with more than 5 tracks in Top 200
Identifies days where five or more distinct tracks from the same artist entered the Top 200 simultaneously . Its a strong, concrete signal of a new album release rather than organic growth. This turns an abstract chart position into a traceable real-world event .

FINDINGS :

The top 3 spots were taken by legendary Danish singer and songwriter Kim Larsen with 50 + tracks charting in a single day on 30th September , 1st and 2nd October . This was because Kim Larsen passed away on 30th September 2018 . Furthermore, out of top 10 Kim Larsen has occupied 7 spots . The rest 3 spots are

occupied by another legendary Swedish Music producer and DJ Avicii with 40+ tracks charting on 3 separate days . Here too his songs blew up and bombarded the charts post his death . Another of such event can be seen where the legendary SoundCloud era rapper XXX tentacion's songs blew up and bombarded onto the charts with 30+ songs charting on the same day post his death . With these concrete evidences we can say posthumous events can be a massive reason for songs to suddenly rise in popularity .

h] 2017 - 2018 Movers :

Compares each artist's total Top 200 appearances year over year, using a LEFT JOIN specifically so that an artist with zero 2017 appearances still appear in the result with a value of zero, rather than being silently excluded. This is what allows the query to correctly surface genuine 2018 breakout artists , those with no meaningful chart presence the year before .

FINDINGS :

XXX Tentacion , Post Malone and Ariana Grande had the biggest number improvements / breakouts in 2018 from 2017 with +59559 , +52026 , +30809 chart entries respectively . Many other prominent Drake , Camilla Cabello and Billie Eilish also had positive figures on charts consistently dominating the tracks .

i] Longest reigning tracks on #1 position

Rather than measuring a track's single best rank ever achieved , this query measures sustained dominance . The total number of individual chart-days a track held rank #1 on Top 200 across all regions in 2018. A song holding the top spot for three consecutive weeks in a major market tells a materially different story than one that touches #1 for a single day before dropping .

FINDINGS :

The top 3 spots are dominated by big players . #1 spot was taken by the song ñTe Boteòby Nio Garcia which spent 1183 days at top spot across all regions but had collaborations by massive latino reggaeton artists like Nicky Jam , Ozuna and Bad Bunny . #2 and #3 spots were taken by none other than by the super famous track óGodô Plan óand óIn my feelings óby Drake which spent 1138 and 1014 days at the top across all the regions .

Data Quality and Limitations :

Stream nulls :

Approximately 2.14 million rows have a null stream value . A direct query confirmed these are 100% Viral 50 entries, matching Spotify's own historical behavior of not publishing raw stream counts for that particular chart.

Combined artist credits :

Collaborations and features are recorded in the source data as a single combined string (for example, "J Balvin, Pharrell Williams, BIA, Sky") rather than as separate individual artists. This means artist-level aggregates reflect credited lineups rather than fully split individual contributions , a known limitation of the source data, left untouched by choice , since splitting it risks false positives .

Track versioning :

A single song can exist under multiple Spotify track IDs ; original version, radio edit, remaster . Each counted separately in this schema. For most tracks this affects only a handful of entries; for a small number of high-profile songs with several official versions charting simultaneously, a track's true aggregate popularity may be split across two or three rows rather than one .

Conclusion :

This project is aimed at treating a single raw file in a very systematic way with a defensible cleaning process . A normalized schema shaped by the actual data without any assumptions

The result is a four-table star schema built entirely from one 3.5GB CSV, capable of answering questions ranging from straightforward chart leaderboards to nuanced ones such as momentum before a peak, hype versus real performance, and a regional expansion timeline .

The next stage of this project connects the completed SQL Server database to Power BI for final visualization, turning these eight queries into an interactive dashboard.